TAI LIEU 2: KHU DE QUY DUYET CAY BANG NGAN XEP (STACK)

Mon hoc: Cau truc du lieu va giai thuat (INT1306) - Bai tap & Nang cao

1. Tai sao dung Stack thay cho De quy?
- Ban chat cua de quy la su dung Call Stack cua he thong. Khi cay qua sau (suy bien O(n)), de quy co nguy co tran bo nho ngan xep (Stack Overflow).
- Ta hoan toan co the dung std::stack<Node*> t??ng minh trên Heap de khu de quy.

2. Code Duyet Pre-order (NLR) bang Stack (Khong de quy):

```
void preOrderIterative(Node* root) {
    if (root == nullptr) return;
    stack<Node*> st;
    st.push(root);
    while (!st.empty()) {
        Node* curr = st.top(); st.pop();
        cout << curr->data << ' ';
        // Day con PHAI truoc, con TRAI sau de con TRAI duoc lay ra truoc (LIFO)
        if (curr->right) st.push(curr->right);
        if (curr->left)  st.push(curr->left);
    }
}
```

DUYET IN-ORDER BANG STACK & BAI TAP TU LUYEN

3. Code Duyet In-order (LNR) bang Stack:

```
void inOrderIterative(Node* root) {
    stack<Node*> st;
    Node* curr = root;
    while (curr != nullptr || !st.empty()) {
        while (curr != nullptr) {
            st.push(curr);
            curr = curr->left; // Di het ve nhanh trai
        }
        curr = st.top(); st.pop();
        cout << curr->data << ' '; // Tham nut
        curr = curr->right; // Chuyen sang nhanh phai
    }
}
```

4. Bai tap thuc hanh:

- Bai 1: Dem so nut la (leaf nodes) tren BST bang Queue (BFS) hoac Stack (DFS).
- Bai 2: Kiem tra mot cay nhi phan co phai la BST hop le hay khong.